

Received 25 June 2026, accepted 27 July 2026, date of publication 10 August 2026, date of current version 24 August 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3722370

RESEARCH ARTICLE

From Pneumonia to Multi-Disease: Interpretable and Uncertainty-Aware Semi-Supervised Learning Strategies for Chest X-Ray Classification

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The work of Jia Uddin and Debasish Ghose was supported in part by UTFORSK Program funded by Norwegian Directorate for Higher Education and Skills (HK-dir) through the Project Sustainable Artificial Intelligence (AI) Literacy in Higher Education through Multilateral Collaborations (SAIL-MC) under Project UTF-2024/10225.

ABSTRACT Pneumonia is a deadly respiratory disease that causes millions of deaths each year worldwide. Chest X-ray imaging is one of the most widely used and affordable tools available for screening pneumonia. However, accurate diagnosis can often be complicated because pneumonia shares similar radiographic features with other respiratory illnesses such as COVID-19 and tuberculosis, leading to potential confusion and misdiagnosis in high-volume clinical environments. To address these issues, this study proposes an automated approach for chest X-ray disease identification by comparing supervised and semi-supervised learning techniques across two publicly available datasets. Experiments were conducted under both binary (Normal vs. Pneumonia) and multiclass (COVID-19, Normal, Pneumonia, Tuberculosis) classification settings. The proposed framework integrates MobileNet-based architectures with modern pseudo-labeling methods, including FixMatch, FlexMatch, and FreeMatch. In supervised experiments, the models achieved up to 98% accuracy in binary classification and 97% in multiclass classification, with MobileNetV3Small selected for its lightweight architecture and deployment efficiency. Under limited labeled data settings, FreeMatch demonstrated the most stable SSL performance, achieving 97% accuracy using only 20% labeled data in binary classification and 30% labeled data in multiclass classification, while maintaining lower variance and improved class balance compared to competing methods. Model interpretability was evaluated using Grad-CAM, confirming that predictions were primarily focused on clinically relevant lung regions. In addition, predictive reliability was assessed using Monte Carlo Dropout, Expected Calibration Error (ECE), and Test-Time Augmentation, demonstrating improved confidence calibration and robustness. Overall, the proposed framework shows that lightweight deep learning models combined with semi-supervised learning, uncertainty estimation, and explainability can provide accurate, efficient, and clinically reliable solutions for automated chest X-ray diagnosis.

INDEX TERMS Pneumonia detection, chest x-ray (CXR), semi-supervised learning, deep learning, MobileNet, FixMatch, FlexMatch, FreeMatch, Grad-CAM, Monte Carlo Dropout.

The associate editor coordinating the review of this manuscript and approving it for publication was Vishal Srivastava.

I. INTRODUCTION

Respiratory diseases remain a major global health concern, causing millions of deaths annually according to global health reports. Among these conditions, pneumonia is one of the leading infectious causes of mortality, particularly among children under five and elderly populations [1]. The disease inflames the air sacs in the lungs, leading to breathing difficulties, severe complications, and potentially fatal outcomes if not diagnosed early [2].

Chest X-ray (CXR) imaging is one of the most commonly used diagnostic tools for identifying lung infections because of its speed, accessibility, and low cost [3], [4]. However, interpreting CXR images requires significant clinical expertise, and even experienced radiologists may misinterpret subtle patterns under heavy workloads. In addition, pneumonia shares several radiographic characteristics with other respiratory infections such as COVID-19 and tuberculosis, which can lead to diagnostic ambiguity in real-world clinical settings [5], [6], [7]. These challenges highlight the need for automated systems capable of accurately identifying pneumonia while effectively distinguishing it from other visually similar lung diseases.

Deep learning has shown remarkable potential in automated CXR analysis. Convolutional neural networks (CNNs) have achieved near-expert performance in detecting disease patterns from medical images [8], [9]. However, most existing models require large labeled datasets, which are costly and time-consuming to obtain. This challenge becomes more significant when attempting to develop reliable diagnostic systems that must differentiate pneumonia from other respiratory diseases with overlapping imaging features.

Semi-supervised learning (SSL) offers an effective solution by leveraging both labeled and unlabeled data during training [10], [11], [12]. Methods such as FixMatch, FlexMatch, and FreeMatch combine pseudo-labeling with consistency regularization to improve model generalization when annotated data is limited. By utilizing large volumes of unlabeled CXRs, SSL enables the development of robust diagnostic models while reducing dependence on labor-intensive manual labeling.

For practical clinical deployment, interpretability and reliability are also essential. Explainable AI (XAI) techniques, such as Gradient-weighted Class Activation Mapping (Grad-CAM), help visualize the regions of the X-ray that influence model predictions, providing clinicians with greater transparency [13], [14]. Additionally, Monte Carlo Dropout can estimate predictive uncertainty, allowing the system to identify cases where model confidence is low and further expert review may be required [15], [16]. Integrating interpretability and uncertainty estimation enhances trust in AI-assisted diagnostic systems.

The individual components have been studied before, but their joint application for chest X-ray diagnosis under limited-label settings is underexplored. Motivated by these challenges, this study focuses primarily on improving pneumonia detection from chest X-ray images. Because pneumonia often exhibits radiographic patterns that resemble

other respiratory infections, models may mistakenly confuse it with diseases such as COVID-19 or tuberculosis. To address this issue, images of these related conditions are also included during training. Their inclusion helps the model better understand the differences between similar lung infections, enabling it to distinguish pneumonia more accurately and reducing the risk of misclassification. The main contributions of this study are summarized as follows:

- Propose a pneumonia-centered multi-disease chest X-ray classification framework incorporating COVID-19 and tuberculosis as clinically related respiratory conditions to enable richer feature learning under a multi-disease learning setting.
- Develop a unified semi-supervised learning pipeline based on FixMatch, FlexMatch, and FreeMatch using a lightweight MobileNetV3-Small backbone for efficient training under limited labeled data.
- Integrate explainability and reliability components, including Grad-CAM visualization and Monte Carlo Dropout-based uncertainty estimation, to enhance interpretability and clinical trust.
- Introduce an uncertainty-aware decision mechanism that flags low-confidence predictions for expert radiologist review, improving diagnostic safety in real-world settings.
- Perform extensive evaluation under imbalanced and low-label settings, demonstrating robust performance with significantly reduced annotation requirements.

The remainder of this paper is organized as follows. Section II reviews existing literature on deep learning-based CXR analysis, semi-supervised learning approaches, and interpretability techniques. Section III presents the proposed methodology, datasets, and experimental setup. Section IV details the evaluation metrics and experimental findings. Section V discusses the implications of the results, and Section VI concludes the study while highlighting potential future directions.

II. RELATED WORKS

A. DEEP LEARNING FOR CHEST X-RAY DISEASE CLASSIFICATION

Deep learning has substantially improved automated chest X-ray (CXR) analysis for respiratory disease diagnosis, particularly for pneumonia, COVID-19, and tuberculosis. One of the earliest landmark studies, CheXNet proposed by Rajpurkar et al. [17], utilized DenseNet121 on the ChestX-ray14 dataset introduced by Wang et al. [18] and achieved radiologist-level pneumonia detection performance. These studies established the foundation for large-scale deep learning-based thoracic disease analysis.

Although early approaches demonstrated strong performance, most focused primarily on binary classification and relied heavily on fully labeled datasets. Such formulations simplify clinical diagnosis and fail to capture the significant visual overlap among pulmonary diseases, particularly pneumonia, COVID-19, and tuberculosis.

To address multiclass diagnosis, several CNN-based frameworks have been proposed. Ozturk et al. [5] developed a DarkNet-based architecture that achieved 98.08% accuracy for binary COVID-19 detection but only 87.02% in multiclass classification involving COVID-19, pneumonia, and normal cases. The large performance drop highlights the difficulty of distinguishing visually similar thoracic abnormalities. COVID-Net [6] introduced an explainable deep learning architecture for COVID-19 detection and incorporated visual attention mechanisms to improve interpretability while maintaining competitive performance.

Recent studies have further improved benchmark accuracy through architectural optimization and preprocessing strategies. Srivastava et al. [7] reported approximately 97.5% accuracy for multiclass respiratory disease classification, while Kumar et al. [19] achieved 98.88% accuracy using image enhancement and feature extraction techniques. Similarly, Alshanketi et al. [20] demonstrated that VGG16 achieved nearly 91% accuracy and outperformed deeper architectures such as ResNet50 and Vision Transformers under limited-data conditions.

Despite these promising results, most existing methods are evaluated on curated or relatively homogeneous datasets and often lack external validation. Consequently, their robustness under domain shifts, noisy labels, and clinically ambiguous multiclass settings remains uncertain. Recent studies have also explored granular computing based feature refinement methods to improve fine-grained image representation prior to deep learning classification [21]. These methods aim to improve discriminative feature learning by multi-level feature aggregation and structured representation of image patterns. Moreover, many frameworks prioritize overall classification accuracy without explicitly addressing pneumonia-centered discrimination against visually overlapping diseases such as COVID-19 and tuberculosis.

B. HYBRID, ENSEMBLE, AND FEDERATED LEARNING FRAMEWORKS

To improve feature representation and classification robustness, hybrid and ensemble learning frameworks have gained increasing attention. Hashmi et al. [22] combined CNN-derived deep features with handcrafted texture descriptors and demonstrated improved pneumonia classification stability. Vyas and Khadatkhar [23] integrated deep learning features with classical machine learning classifiers such as decision trees and random forests, reporting accuracies approaching 99.98% on curated datasets.

Advanced multiclass architectures have also demonstrated strong benchmark performance. Malik et al. [24] proposed CDC_Net, a residual and dilated convolution-based framework for five-class chest disease classification, achieving 99.39% accuracy. However, the framework remains fully supervised and does not address uncertainty estimation, annotation scarcity, or reliability-aware learning.

Federated learning approaches have further expanded privacy-preserving medical AI. DMFL_Net [25] achieved

98.45% accuracy for multiclass chest disease classification using distributed federated learning. Similarly, Malik et al. [26] proposed a blockchain-assisted federated learning framework integrating Capsule Networks and incremental extreme learning machines for COVID-19 CT classification. Although these approaches improve decentralized learning and patient privacy preservation, they still depend heavily on fully annotated datasets and do not incorporate explainability-aware or uncertainty-aware modeling.

Overall, existing hybrid and federated frameworks primarily emphasize benchmark accuracy while providing limited focus on low-label learning, uncertainty estimation, and clinically reliable multiclass discrimination.

C. EXPLAINABILITY AND UNCERTAINTY IN CLINICAL AI

Interpretability is a fundamental requirement for clinical AI systems because medical diagnosis requires transparent and trustworthy decision-making. Gradient-based visualization techniques such as Grad-CAM and Grad-CAM++ [14], [27] are widely used to highlight disease-relevant regions in chest X-rays. COVID-Net [6] further demonstrated that integrating explainability into deep learning pipelines can improve clinician interpretability without significantly sacrificing classification performance.

In parallel, Monte Carlo Dropout [15] has emerged as an effective Bayesian approximation method for predictive uncertainty estimation. Recent medical imaging studies demonstrated that uncertainty estimation can identify unreliable or out-of-distribution predictions and improve decision reliability [16], [28].

However, most existing frameworks still treat explainability and uncertainty estimation as independent post hoc components rather than integrating them into a unified reliability-aware diagnostic framework. Consequently, many systems remain difficult to trust in clinically ambiguous scenarios where prediction confidence is critical.

D. SEMI-SUPERVISED LEARNING IN MEDICAL IMAGING

Semi-supervised learning (SSL) has gained considerable attention because medical image annotation is expensive, time-consuming, and dependent on expert radiologists. SSL frameworks leverage unlabeled data through pseudo-labeling and consistency regularization, reducing reliance on fully labeled datasets.

FixMatch [10] introduced confidence-based pseudo-labeling with weak and strong augmentation strategies and demonstrated strong low-label learning performance. FlexMatch [11] further improved pseudo-label reliability through curriculum learning, while FreeMatch [12] introduced adaptive thresholding mechanisms for enhanced training stability.

Although SSL methods achieve strong performance on benchmark datasets, most evaluations remain limited to natural image datasets. In medical imaging, particularly multiclass respiratory disease classification, SSL remains

relatively underexplored. Furthermore, existing SSL frameworks rarely integrate explainability and uncertainty estimation into the learning pipeline.

Another major limitation is that most SSL approaches are not explicitly designed to address diagnostic ambiguity among visually similar pulmonary diseases. Incorrect pseudo-label generation in multiclass settings can propagate errors during training and reduce model reliability.

E. META-ANALYSIS AND RESEARCH GAPS

Table 1 presents representative studies selected to summarize key methodological directions and performance trends in chest X-ray based deep learning.

TABLE 1. Representative meta-analysis of chest X-ray deep learning studies.

Study	Dataset	Method	Result	Limitation
CheXNet [17]	CXR14	DenseNet121 TL	F1=0.435	Weak labels, limited external validation
COVID-Net [6]	COVIDx	CNN (PEPX)	93.3% acc	Imbalance, dataset bias
Ozturk et al. [5]	COVID-19 Image Data Collection database + ChestX-ray8 database	DarkCovidNet (DarkNet-based CNN)	98.08% / 87.02%	Small and heterogeneous COVID-19 database, class imbalance, and limited external validation
Hashmi et al. [22]	CXR dataset	CNN + hand-crafted features	Improved stability	Limited generalization reported
Vyas et al. [23]	CXR dataset	ML, DL with stacking	99.9% acc	Limited dataset diversity, no external validation, possible overfitting, weak real-world generalization
Srivastava et al. [7]	CXR	VGG-19 TL	97.5% acc	Class imbalance and potential dataset-specific bias due to heterogeneous data sources
Kumar et al. [19]	CXR datasets	CNN + enhancement	98.88% acc	Limited real-world validation
Malik et al. [24]	Multi-class CXR	Residual CNN (CDC-Net)	99.39% acc	Real-world clinical testing

The literature reveals several important research gaps. First, most high-performing CXR frameworks rely heavily on fully supervised learning and curated datasets, limiting applicability in low-label and clinically heterogeneous environments [17], [24], [25]. Second, although SSL methods improve low-label learning performance, they are rarely validated for multiclass respiratory disease classification involving strong inter-class similarity [10], [11], [12]. Third, explainability and uncertainty estimation are generally treated as isolated post hoc components rather than integrated into a unified reliability-aware diagnostic framework [6], [15], [16], [27].

Most importantly, relatively few studies explicitly focus on pneumonia-centered discrimination against visually overlapping diseases such as COVID-19 and tuberculosis, despite

this being one of the most clinically challenging aspects of thoracic diagnosis [5], [6], [24]. Therefore, a unified framework integrating semi-supervised learning, explainability, and uncertainty estimation remains necessary for robust and clinically trustworthy pneumonia detection.

III. METHODOLOGY

This study proposes a hybrid deep learning framework for chest X-ray analysis that integrates supervised and semi-supervised learning strategies to improve pneumonia detection, particularly in scenarios with limited labeled data. To enhance diagnostic reliability, additional respiratory conditions such as COVID-19 and tuberculosis are included during training, enabling the model to differentiate pneumonia from visually similar diseases.

The supervised component employs MobileNet variants for efficient feature extraction, making the framework suitable for deployment in resource-constrained clinical environments. The semi-supervised component leverages unlabeled images using FixMatch, FlexMatch, and FreeMatch strategies, with weak and strong data augmentations applied to enforce consistency and improve generalization.

By combining these pipelines, the proposed framework provides robust pneumonia detection while maintaining high differentiation from other respiratory infections. The overall workflow of the framework is illustrated in Figure 1.

A. DATASET PREPARATION

This study utilizes two publicly available chest X-ray datasets to evaluate pneumonia classification under binary and multiclass settings. Both datasets are fully anonymized, with all patient-identifying information removed to ensure compliance with ethical standards and patient privacy protection. The first dataset, the *Chest X-Ray Images (Pneumonia) Dataset* [3], is hosted on Mendeley Data and contains 5,863 pediatric chest X-ray images collected from retrospective clinical cohorts at the Guangzhou Women and Children's Medical Center. The dataset consists of two classes: Normal and Pneumonia. It is organized into training, validation, and test splits and was originally curated for automated pneumonia detection in pediatric patients. The diagnostic labels were verified by multiple expert physicians to ensure high annotation reliability, making it a widely adopted benchmark for medical image classification. The second dataset, the *Chest X-Ray Pneumonia, COVID-19, and Tuberculosis Dataset* [4], is a merged multi-source dataset containing 7,135 chest X-ray images across four categories: COVID-19, Normal, Pneumonia, and Tuberculosis. The dataset is provided with predefined training, validation, and test splits in the Kaggle repository. The training set contains 6,326 images, while the test set contains 771 images and the validation set contains 38 images. This dataset integrates multiple publicly available repositories, including the Mendeley Chest X-Ray Pneumonia dataset [3], COVID-19 chest X-ray collections from open repositories [29], [30], and tuberculosis datasets [31]. As a result, the multiclass dataset extends the binary dataset by incorporating additional respiratory

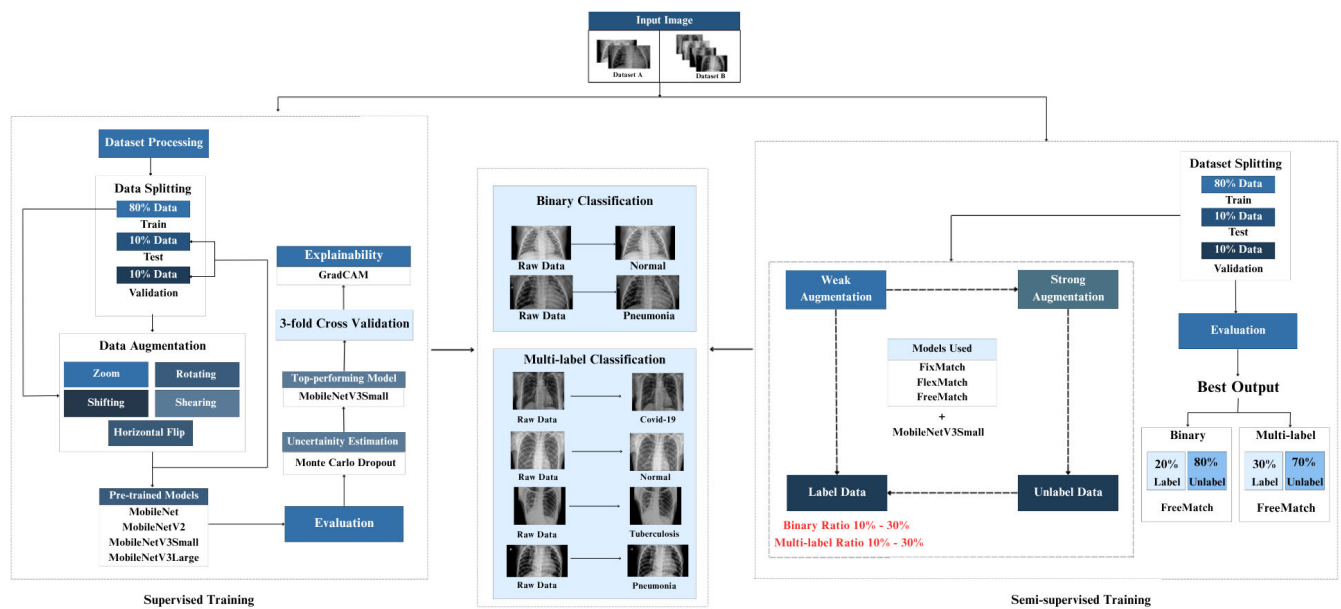


FIGURE 1. Workflow of the proposed semi-supervised pneumonia classification framework.

diseases, while also introducing partial overlap in underlying sources. Although the multiclass dataset includes images from the same Mendeley pneumonia repository used in the binary task, both settings were constructed with independent train, validation, and test splits. The binary and multiclass experiments address different classification objectives, and no cross-task overlap or evaluation was performed, ensuring no information leakage between tasks. Experiments were conducted in two settings:

- Binary classification: Normal vs. Pneumonia using the Mendeley pediatric chest X-ray dataset.
- Multiclass classification: COVID-19, Normal, Pneumonia, and Tuberculosis using the combined multi-source dataset.

The datasets exhibit significant class imbalance, primarily due to the dominance of Pneumonia cases. To mitigate this issue, controlled sampling strategies were applied to construct balanced and computationally efficient training sets while preserving clinically meaningful radiographic variability.

1) BINARY DATASET

The binary classification experiments were conducted using the pediatric chest X-ray dataset from Mendeley Data [3]. The dataset consists of two categories: Normal and Pneumonia. The dataset exhibits substantial class imbalance due to the dominance of pneumonia cases. To construct a computationally balanced experimental setting while reducing class

dominance bias, a controlled subset of 1300 images was randomly selected from each class. The resulting balanced dataset distribution is shown in Table 2.

TABLE 2. Binary dataset distribution after controlled sampling.

Class	Original Images	Selected Images
Normal	1341	1300
Pneumonia	3875	1300
Total	5216	2600

In addition to the balanced setting, experiments were also performed on the original imbalanced dataset to evaluate model behavior under naturally occurring clinical class distributions. Representative chest X-ray samples from the binary dataset are shown in Figure 2.



(a) Normal



(b) Pneumonia

FIGURE 2. Representative chest X-ray samples from the binary classification dataset.

2) MULTICLASS DATASET

The multiclass classification experiments were conducted using the *Chest X-Ray Pneumonia, COVID-19, and*

Tuberculosis Dataset [4]. This dataset combines chest X-ray images from multiple public repositories and contains four respiratory disease categories. The dataset is highly imbalanced, with Pneumonia being the dominant class. Such imbalance can bias deep learning models toward majority classes and reduce sensitivity for minority disease categories. To address this issue, a two-stage strategy was adopted rather than aggressive undersampling to a single minority class size. In the first stage, controlled sampling was applied to reduce the dominant Pneumonia class from 3,875 to 1,300 images consistent with the size of the Normal class in the original binary dataset, while retaining all COVID-19 (460) and Tuberculosis (650) images to preserve clinically meaningful radiographic diversity. In the second stage, offline oversampling was applied to the training set to expand all classes to 1,140 images, achieving full class balance without discarding useful data. To address class imbalance, a controlled offline random oversampling strategy was applied. No synthetic sample generation was used, and oversampling was performed strictly after train-validation-test splitting to avoid data leakage.

This two-stage approach avoids the excessive data loss associated with undersampling to the smallest minority class (460 images) and reduces the generalization risk identified in prior work. The resulting dataset distribution after controlled sampling and oversampling is summarized in Table 3.

TABLE 3. Multiclass dataset distribution after controlled sampling and oversampling.

Class	Original Images	Selected Images	After Oversampling
COVID-19	460	460	1140
Normal	1341	1300	1140
Pneumonia	3875	1300	1140
Tuberculosis	650	650	1140
Total	6326	3710	4560

Representative multiclass chest X-ray samples are shown in Figure 3.

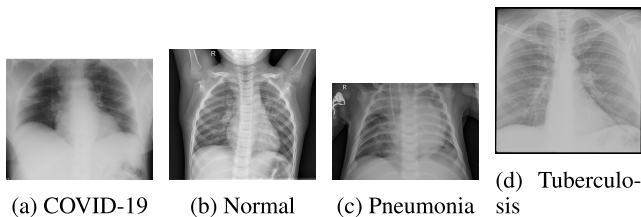


FIGURE 3. Representative chest X-ray samples from the multiclass dataset.

3) DATASET SPLITTING

Stratified sampling was employed to preserve class distribution across training, validation, and testing subsets. Dataset splitting was performed prior to augmentation to prevent data leakage between subsets. To ensure experimental reproducibility, a fixed random seed ($SEED = 42$) was used for all stochastic operations, including dataset shuffling, controlled sampling, train-validation-test splitting,

and TensorFlow-based model initialization. For the binary dataset, the balanced dataset was divided into:

- Training set: 2080 images
- Validation set: 260 images
- Test set: 260 images

For the multiclass dataset, a hybrid per-class sampling strategy was adopted. Specifically, 80 images per class were randomly selected for validation and another 80 images per class were reserved for testing to ensure balanced evaluation across disease categories. Although this design ensures balanced evaluation, the relatively small validation size (80 images per class) may introduce variance in performance estimates; however, multiple runs and stratified sampling were used to mitigate this effect and maintain stable SSL and cross-validation results. The final multiclass split distribution is presented in Table 4.

TABLE 4. Final multiclass dataset split distribution.

Class	Training	Validation	Testing
COVID-19	300	80	80
Normal	1140	80	80
Pneumonia	1140	80	80
Tuberculosis	490	80	80
Total	3070	320	320

Additionally, cross-validation was applied in the supervised learning setting to ensure robust model evaluation and reduce variance in performance estimates.

4) LABEL ENCODING

All chest X-ray categories were converted into numerical labels to ensure compatibility with deep learning classification models.

Binary classification labels:

- NORMAL $\rightarrow 0$
- PNEUMONIA $\rightarrow 1$

Multiclass classification labels:

- COVID-19 $\rightarrow 0$
- NORMAL $\rightarrow 1$
- PNEUMONIA $\rightarrow 2$
- TUBERCULOSIS $\rightarrow 3$

Each image path was paired with its corresponding encoded label to construct structured dataframes for training and evaluation.

B. DATA PREPROCESSING AND AUGMENTATION

1) IMAGE PREPROCESSING

All chest X-ray images were resized to 224×224 pixels prior to training to ensure compatibility with MobileNet-based architectures. Since the pretrained MobileNet models were originally trained on ImageNet, grayscale chest X-ray images were converted into three-channel RGB format. All images were subsequently normalized using the MobileNetV3 `preprocess_input` function. For validation and testing datasets, only preprocessing operations were applied without augmentation to ensure unbiased evaluation using original clinical images.

2) ONLINE AUGMENTATION

To improve model robustness and reduce overfitting, online augmentation was applied during training using the `ImageDataGenerator` module from Keras. The augmentation pipeline simulates realistic variations in chest X-ray acquisition conditions, including minor patient positioning differences and imaging inconsistencies. The augmentation configuration is summarized in Table 5.

TABLE 5. Online data augmentation configuration.

Augmentation Technique	Parameter Value
Rotation Range	$\pm 15^\circ$
Width Shift Range	0.05
Height Shift Range	0.05
Zoom Range	0.10
Shear Range	0.05
Horizontal Flip	No
Vertical Flip	No
Fill Mode	Nearest

Images were randomly rotated, shifted, zoomed, and sheared to improve generalization capability under varying acquisition conditions. Horizontal and vertical flipping were disabled to preserve anatomical orientation and avoid unrealistic transformations in chest radiographs.

C. MODEL ARCHITECTURE

This study investigates both supervised and semi-supervised chest X-ray classification using lightweight transfer learning architectures from the MobileNet family. The selected models include MobileNet [9], MobileNetV2 [32], MobileNetV3Small, and MobileNetV3Large [33], all initialized with ImageNet pretrained weights.

The MobileNet family was selected due to its favorable balance between computational efficiency and classification performance. Lightweight architectures are particularly suitable for medical imaging applications deployed in resource-constrained clinical environments such as portable radiography systems, low-cost diagnostic devices, and edge-based healthcare platforms.

1) MOBILENET FAMILY

MobileNet utilizes depthwise separable convolutions to significantly reduce parameter count and computational complexity while preserving representational capability. MobileNetV2 introduces inverted residual bottleneck blocks and linear bottlenecks, enabling improved feature propagation and stable fine-tuning performance. MobileNetV3 further improves efficiency through neural architecture search and squeeze-and-excitation attention modules.

Among the evaluated variants, MobileNetV3Small prioritizes lower computational overhead and faster inference, whereas MobileNetV3Large provides greater representational capacity for extracting complex radiographic patterns from chest X-ray images.

All architectures were adapted for binary and multiclass respiratory disease classification by replacing the original

ImageNet classification head with a task-specific fully connected classifier.

2) SUPERVISED LEARNING ARCHITECTURE

For supervised learning experiments, the pretrained backbone network was combined with a custom classification head designed for chest X-ray analysis. The architecture includes global average pooling, fully connected layers, batch normalization, and dropout regularization to improve feature generalization and reduce overfitting. The final output layer used sigmoid activation for binary classification and softmax activation for multiclass classification. Transfer learning was applied by freezing most pretrained layers during early training while fine-tuning the deeper convolutional blocks to adapt the model to radiographic image characteristics. The configuration of the supervised classification model is summarized in Table 6.

TABLE 6. Supervised model architecture configuration.

Component	Configuration
Backbone Networks	MobileNet, MobileNetV2, MobileNetV3
Pretrained Weights	ImageNet
Input Resolution	$224 \times 224 \times 3$
Frozen Layers	All except last 20 layers
Pooling Layer	Global Average Pooling
Dense Layers	$512 \rightarrow 256$ neurons
Batch Normalization	Applied after dense layers
Dropout Rates	0.5 and 0.4
Binary Output Activation	Sigmoid
Multiclass Output Activation	Softmax
Optimizer	Adam
Initial Learning Rate	3×10^{-4}
Learning Rate Strategy	Cosine decay
Loss Function	Binary / Sparse Categorical Cross-Entropy

Global average pooling was employed to reduce spatial dimensionality while preserving discriminative semantic information. Batch normalization improved optimization stability, whereas dropout regularization reduced overfitting caused by limited medical imaging samples.

The final classification layers were intentionally lightweight to maintain computational efficiency while enabling effective adaptation to chest radiography datasets.

D. SEMI-SUPERVISED LEARNING FRAMEWORK

Although supervised learning achieves strong performance when sufficient annotations are available, medical imaging datasets are often limited by expensive and time-consuming expert labeling procedures. To address this challenge, semi-supervised learning (SSL) was incorporated to utilize both labeled and unlabeled chest X-ray images.

The SSL framework combines:

- Supervised learning on labeled images
- Pseudo-label generation on unlabeled images
- Consistency regularization using weak and strong augmentations

Three state-of-the-art consistency-based SSL approaches were evaluated: FixMatch [10], FlexMatch [11] and FreeMatch [12]

1) LABELED AND UNLABELED DATA PARTITIONING

To simulate realistic low-label clinical environments, the balanced multiclass training dataset was partitioned into labeled and unlabeled subsets using stratified sampling. Three labeling scenarios were considered. The remaining samples in each setting were treated as unlabeled during training by removing their ground-truth annotations. The dataset-level SSL partitioning is summarized in Table 7.

TABLE 7. Labeled and unlabeled partitioning.

Dataset	Labeled Ratio	Labeled Images	Unlabeled Images	Total
Binary	10%	208	1872	2080
	20%	416	1664	2080
	30%	624	1456	2080
Multiclass	10%	456	4104	4560
	20%	912	3648	4560
	30%	1368	3192	4560

Class-wise stratification was preserved across all subsets to maintain balanced disease representation.

2) SEMI-SUPERVISED MODEL ARCHITECTURE

The same MobileNet-based backbone used in supervised learning was also employed for SSL experiments to ensure fair comparison between supervised and semi-supervised paradigms. However, during SSL training, the network receives two separate image streams: labeled and unlabeled chest X-ray images.

3) WEAK AND STRONG AUGMENTATION STRATEGY

Weak and strong augmentation strategies were incorporated into the semi-supervised learning (SSL) framework to improve model generalization and consistency learning under limited labeled supervision.

a: WEAK AUGMENTATION

Weak augmentation applies mild geometric perturbations while preserving anatomical structures. It is used for supervised training and pseudo-label generation.

The weak augmentation configuration is summarized in Table 8.

TABLE 8. Weak augmentation parameters.

Preprocessing	MobileNetV3 normalization
Rotation Range	$\pm 8^\circ$
Width Shift	0.05
Height Shift	0.05
Zoom Range	0.08
Horizontal Flip	Disabled
Fill Mode	Nearest

Representative weakly augmented chest X-ray samples are shown in Figure 4.

b: STRONG AUGMENTATION

The strong augmentation pipeline includes geometric transformations, contrast enhancement, brightness variations, Gaussian blur, and random cropping. The strong augmentation pipeline includes geometric transformations, contrast

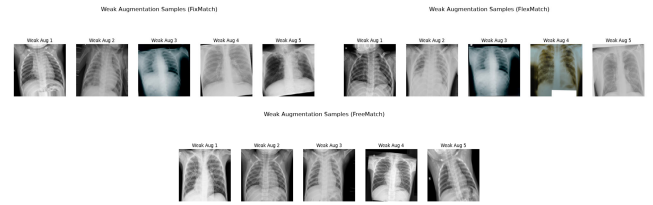


FIGURE 4. Weak augmentations used in semi-supervised learning.

enhancement, brightness variations, Gaussian blur, and random cropping. The configuration is summarized in Table 9.

TABLE 9. Strong augmentation parameters.

Rotation	$\pm 8^\circ$ ($p = 0.5$)
ShiftScaleRotate	shift=0.05, scale=0.05 ($p = 0.5$)
Brightness/Contrast	$p = 0.6$
CLAHE	$p = 0.5$
Gaussian Blur	limit=3 ($p = 0.2$)
Random Resized Crop	scale 0.9–1.0 ($p = 0.5$)

Representative strongly augmented images are shown in Figure 5.

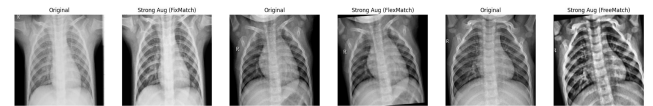


FIGURE 5. Strong augmentation examples applied to unlabeled chest X-ray images during SSL training.

4) PSEUDO-LABEL GENERATION

For each unlabeled sample $u \in D_u$, pseudo-labels are generated using predictions from weakly augmented inputs. Let $f_\theta(\cdot)$ denote the neural network parameterized by θ , and let $u_w = a_w(u)$ represent the weakly augmented image.

The predictive probability distribution is defined as:

$$p_\theta(u_w) = f_\theta(u_w) \quad (1)$$

The pseudo-label is obtained using the maximum predicted probability:

$$\hat{y} = \arg \max_k p_\theta^{(k)}(u_w) \quad (2)$$

The associated confidence score is:

$$p_{\max} = \max_k p_\theta^{(k)}(u_w) \quad (3)$$

A confidence-based binary mask is then applied:

$$\mathcal{M} = \mathbb{I}(p_{\max} \geq \tau) \quad (4)$$

where τ denotes the confidence threshold. Depending on the SSL framework, τ is fixed (FixMatch), class-adaptive (FlexMatch), or dynamically updated (FreeMatch). Only samples satisfying $\mathcal{M} = 1$ contribute to the unsupervised objective, thereby reducing noisy pseudo-label supervision and confirmation bias.

5) FIXMATCH METHOD

FixMatch employs a fixed confidence threshold $\tau = 0.90$ for pseudo-label filtering.

The unsupervised objective is defined as:

$$\mathcal{L}_{unsup} = \mathbb{I}(p_{\max} \geq 0.90) \cdot CE(\hat{y}, f_{\theta}(u_s)) \quad (5)$$

where $u_s = a_s(u)$ denotes the strongly augmented unlabeled sample.

6) FLEXMATCH METHOD

FlexMatch extends FixMatch by introducing class-wise adaptive confidence thresholds to better handle class imbalance and varying learning difficulty across disease categories.

Each class c maintains a dynamic threshold τ_c , updated using exponential moving average (EMA):

$$\tau_c \leftarrow \alpha \tau_c + (1 - \alpha) \mu_c \quad (6)$$

where μ_c denotes the mean confidence of pseudo-label predictions belonging to class c .

To improve prediction stability, FlexMatch additionally enforces consistency between multiple weak augmentations:

$$f_{\theta}(a_w(u)) \approx f_{\theta}(a'_w(u)) \quad (7)$$

Furthermore, a top-confidence selection strategy is applied to retain only the most reliable pseudo-labels during unsupervised training, thereby reducing confirmation bias and training instability.

7) FreeMatch METHOD

FreeMatch extends FixMatch by eliminating manually fixed thresholds and introducing adaptive distribution-aware pseudo-label selection.

During SSL training, a global confidence threshold τ is dynamically updated using exponential moving average (EMA):

$$\tau \leftarrow m\tau + (1 - m)\mathbb{E}[p_{\max}] \quad (8)$$

where m represents the EMA momentum coefficient and $\mathbb{E}[p_{\max}]$ denotes the mean pseudo-label confidence within the current batch.

To model class-wise prediction dynamics, a running class distribution estimator is maintained:

$$p_{model} \leftarrow mp_{model} + (1 - m)\bar{p} \quad (9)$$

where $\bar{p}$ is the batch-wise mean predicted probability distribution.

For each unlabeled sample u_i , the class-aware adaptive threshold is computed as:

$$\tau_i^{adaptive} = \tau \cdot p_{model}(c_i) \quad (10)$$

where:

$$c_i = \arg \max_k p_{\theta}^{(k)}(u_i) \quad (11)$$

The corresponding selection mask is defined as:

$$\mathcal{M}_i = \mathbb{I}(p_{\max,i} \geq \tau_i^{adaptive}) \quad (12)$$

Only selected samples contribute to the unsupervised objective:

$$\mathcal{L}_{unsup} = \frac{1}{|\mathcal{M}|} \sum_{i: \mathcal{M}_i=1} CE(\hat{y}_i, f_{\theta}(a_s(u_i))) \quad (13)$$

To prevent class collapse and encourage balanced predictions, a fairness regularization term is introduced:

$$\mathcal{L}_{fair} = \left\| \bar{p} - \frac{1}{C} \right\|^2 \quad (14)$$

where C denotes the total number of classes.

The final optimization objective is:

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda_u \mathcal{L}_{unsup} + 0.01 \mathcal{L}_{fair} \quad (15)$$

The unsupervised weight is gradually increased during training according to:

$$\lambda_u = \lambda_{max} \frac{t}{T} \quad (16)$$

where t and T denote the current and total SSL epochs, respectively.

8) TRAINING PROCEDURE

Training was performed in two sequential stages:

1) Warm-up Phase:

All SSL methods begin with supervised warm-up training using labeled data only for $E_w = 3$ epochs. During this stage, categorical cross-entropy loss is minimized:

$$\mathcal{L}_{sup} = - \sum_{c=1}^C y_c \log(p_c) \quad (17)$$

where C denotes the total number of classes and p_c represents the predicted probability for class c . This phase stabilizes feature learning before introducing pseudo-label supervision.

2) Semi-Supervised Phase:

Following warm-up, labeled and unlabeled samples are jointly optimized using consistency regularization and confidence-filtered pseudo-label supervision.

The best-performing model checkpoint was selected based on validation accuracy.

TABLE 10. Training hyperparameters.

Optimizer	Adam
Learning Rate	3×10^{-4}
Batch Size	32
Warm-up Epochs	3
SSL Epochs	5
Number of Runs	3
Pseudo-label Threshold	0.90
Backbone Network	MobileNetV3Small
Input Resolution	224×224

9) TRAINING CONFIGURATION

All models were optimized using the Adam optimizer with learning rate 3×10^{-4} . Training was conducted for five SSL epochs following supervised warm-up. Each iteration jointly processes labeled and unlabeled mini-batches. (Table 10)

10) ALGORITHMIC FRAMEWORK

The proposed semi-supervised learning framework follows a unified FixMatch-style pipeline consistently applied across FixMatch, FlexMatch, and FreeMatch variants. The primary difference among methods lies in the pseudo-label thresholding strategy, while the overall optimization framework remains unchanged.

Algorithm 1 summarizes the unified SSL training procedure.

Algorithm 1 Unified SSL Framework for FixMatch, FlexMatch, and FreeMatch

Require: Labeled data D_l , unlabeled data D_u

```

1: for each iteration do
2:   Sample  $(X_l, Y_l) \sim D_l, X_u \sim D_u$ 
3:    $\mathcal{L}_{sup} = CE(f_\theta(X_l), Y_l)$ 
4:    $p_u = f_\theta(a_w(X_u))$ 
5:    $\hat{Y}_u = \arg \max(p_u), p_{\max} = \max(p_u)$ 
6:   if FixMatch then
7:      $M = \mathbb{I}(p_{\max} \geq 0.90)$ 
8:   end if
9:   if FlexMatch then
10:     $M = \mathbb{I}(p_{\max} \geq \tau_c)$ 
11:    Apply consistency filtering
12:   end if
13:   if FreeMatch then
14:     Update EMA threshold  $\tau$ 
15:      $\tau_i^{adaptive} = \tau \cdot p_{model}(c_i)$ 
16:      $M = \mathbb{I}(p_{\max} \geq \tau_i^{adaptive})$ 
17:   end if
18:    $X_u^s = a_s(X_u[M])$ 
19:    $\mathcal{L}_{unsup} = CE(f_\theta(X_u^s), \hat{Y}_u[M])$ 
20:    $\mathcal{L} = \mathcal{L}_{sup} + \lambda_u \mathcal{L}_{unsup}$ 
21:   Update  $\theta$ 
22: end for

```

Overall, the proposed framework effectively leverages unlabeled chest X-ray images through confidence-based pseudo-label filtering and consistency regularization. FixMatch employs a fixed thresholding strategy, FlexMatch introduces class-adaptive confidence thresholds, and FreeMatch further extends the framework using dynamic distribution-aware threshold estimation.

E. MODEL EVALUATION AND RELIABILITY FRAMEWORK

1) CLASSIFICATION METRICS

Model performance was evaluated using standard metrics derived from the confusion matrix, where TP , TN , FP , and FN denote true positives, true negatives, false positives,

and false negatives, respectively. The evaluation metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics provide complementary perspectives on classification performance. Accuracy reflects overall correctness, precision measures the reliability of positive predictions, and recall captures sensitivity to positive cases. The F1-score balances precision and recall, making it particularly useful under class imbalance.

For multiclass evaluation, macro-averaged and weighted-averaged versions of these metrics were also computed to account for residual class imbalance.

2) CROSS-VALIDATION STRATEGY

To improve robustness and reduce variance from a single train-validation split, a 3-fold stratified cross-validation strategy was employed using StratifiedKFold from Scikit-learn with shuffling enabled and a fixed random seed (`random_state = 42`) for reproducibility.

Given dataset D , stratified cross-validation partitions it into $K = 3$ folds: $D = \{D_1, D_2, D_3\}$, where each fold preserves the original class distribution: $P_c(D_i) \approx P_c(D)$.

In each iteration, one fold is used for validation while the remaining folds are used for training, i.e., $D_{train} = D \setminus D_i$, $D_{val} = D_i$.

The final performance is reported as the average across folds: $\bar{M} = \frac{1}{K} \sum_{i=1}^K M_i$, where M_i is the metric for fold i .

Stratification ensures that each fold maintains representative class distributions, which is particularly important for imbalanced chest X-ray datasets.

3) EXPLAINABILITY AND RELIABILITY ANALYSIS

To improve interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was used to highlight regions contributing to model predictions. The method generates a class-specific heatmap by weighting feature maps from the convolutional layers of the network, specifically intermediate activations 13 and deeper activations 17 of the MobileNetV3 backbone, enabling analysis of feature importance at different semantic depths:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right), \quad \alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

Here, A^k represents feature maps extracted from selected convolutional layers, while α_k^c denotes the importance of each feature map for class c based on gradient information. The resulting heatmaps were overlaid on chest X-ray images to verify clinically meaningful regions and to assess whether mid-level or deeper convolutional representations focus on pathology-relevant areas.

4) MONTE CARLO DROPOUT FOR UNCERTAINTY ESTIMATION

Predictive uncertainty was estimated using Monte Carlo Dropout (MC Dropout), where dropout remains active during inference to simulate multiple stochastic forward passes. This allows approximation of Bayesian uncertainty without modifying the model. The procedure is summarized in Algorithm 2.

Algorithm 2 Monte Carlo Dropout Inference

Require: Trained model f_θ , input image x , stochastic passes T

Ensure: Mean prediction $\bar{p}$ and uncertainty H

- 1: Initialize $\mathcal{P} \leftarrow []$
- 2: **for** $t = 1$ to T **do**
- 3: Enable dropout during inference
- 4: Generate prediction $p_t \leftarrow f_\theta(x, \text{training}=\text{True})$
- 5: Store p_t in $\mathcal{P}$
- 6: **end for**
- 7: Compute mean prediction:

$$\bar{p} = \frac{1}{T} \sum_{t=1}^T p_t$$

- 8: Compute predictive uncertainty using entropy:

$$H(p) = - \sum_{c=1}^C \bar{p}_c \log(\bar{p}_c + \epsilon)$$

- 9: **return** $\bar{p}, H$
-

The mean prediction $\bar{p}$ represents the final output probability, while entropy $H(p)$ measures uncertainty. Higher entropy indicates inconsistent predictions across stochastic passes, which often corresponds to ambiguous or low-quality radiographs. An entropy threshold of $\tau = 0.30$ was used to flag uncertain cases for expert review. The threshold was selected empirically based on the distribution of predictive entropy values obtained from Monte Carlo Dropout inference using 50 stochastic forward passes. This value provided a practical operating point that balances reliable prediction coverage with uncertainty-based referral.

5) TEST-TIME AUGMENTATION AND CALIBRATION ANALYSIS

To improve robustness, Test-Time Augmentation (TTA) was applied by generating multiple augmented versions of each input image and averaging their predictions:

$$\bar{p}_{TTA} = \frac{1}{K} \sum_{k=1}^K p_k, \quad \hat{y} = \arg \max(\bar{p}_{TTA})$$

This reduces sensitivity to minor variations in image acquisition and improves prediction stability.

Finally, model reliability was evaluated using Expected Calibration Error (ECE), which measures the alignment

between predicted confidence and actual accuracy:

$$ECE = \sum_{m=1}^M \frac{n_m}{N} |\text{acc}(m) - \text{conf}(m)|$$

Lower ECE indicates better-calibrated probabilistic outputs, which is critical for clinical decision support systems.

IV. RESULT ANALYSIS

A. SUPERVISED LEARNING PERFORMANCE

The proposed MobileNet-based models were evaluated under a supervised learning setup on held-out chest X-ray test data. Two classification scenarios were considered: binary (Normal vs. Pneumonia) and multiclass (COVID-19, Normal, Pneumonia, Tuberculosis). The binary task emphasizes pneumonia detection, while the multiclass setting ensures the model can differentiate pneumonia from other respiratory conditions with similar radiographic patterns. Performance was assessed using accuracy, precision, recall, F1-score, as well as deployment efficiency, cross-validation, interpretability, uncertainty, and semi-supervised learning experiments.

1) BINARY AND MULTICLASS RESULTS

Table 11 and Table 12 summarize the classification performance.

TABLE 11. Binary classification performance of mobilenet models.

Model	Accuracy	Precision	Recall	F1-score
MobileNet	0.98	0.99	0.98	0.98
MobileNetV2	0.97	0.97	0.97	0.97
MobileNetV3Small	0.97	0.97	0.97	0.97
MobileNetV3Large	0.95	0.95	0.95	0.95

TABLE 12. Multiclass classification performance of mobilenet models.

Model	Accuracy	Precision	Recall	F1-score
MobileNet	0.97	0.97	0.97	0.97
MobileNetV3Small	0.95	0.95	0.95	0.95
MobileNetV3Large	0.90	0.93	0.90	0.90
MobileNetV2	0.79	0.85	0.79	0.78

MobileNet and MobileNetV3Small consistently achieved the highest accuracy across both tasks, with MobileNet attaining 98% in the binary task and 97% in multiclass classification. MobileNetV2 and MobileNetV3Large showed slightly lower performance, particularly in the multiclass setting, likely due to misclassification among visually similar conditions. These results confirm that lightweight MobileNet variants are effective for chest X-ray classification while remaining computationally efficient.

2) DEPLOYMENT EFFICIENCY ANALYSIS

In addition to classification accuracy, deployment efficiency was evaluated using three metrics: Floating Point Operations (FLOPs), CPU memory consumption, and inference latency. These metrics are important in clinical environments where AI models may need to operate on resource-constrained systems such as portable X-ray machines or edge devices.

MobileNet and MobileNetV3Small were selected for deployment-oriented evaluation because they achieved the strongest overall classification performance among the investigated architectures. MobileNet produced the highest predictive accuracy in both binary and multiclass tasks, while MobileNetV3Small achieved competitive performance with substantially lower computational requirements. This combination enables a meaningful comparison between maximum predictive performance and lightweight real-time deployment efficiency.

TABLE 13. Deployment efficiency comparison of mobilenet models.

Model	FLOPs	Memory Usage	Inference Time
MobileNet	1147M	2048 MB	0.23 s
MobileNetV3Small	116.9M	1575 MB	0.11 s

Table 13 summarizes the deployment efficiency metrics, showing that MobileNetV3Small requires approximately ten times fewer floating point operations (116.9M vs. 1147M FLOPs), indicating substantially lower computational complexity.

In addition, MobileNetV3Small demonstrates reduced memory consumption and faster inference speed. The inference latency decreases from 0.23 seconds for MobileNet to 0.11 seconds for MobileNetV3Small, representing nearly a 50% improvement in prediction time. Such improvements are particularly valuable for real-time clinical decision support systems where rapid processing and limited computational resources are critical.

Although MobileNet achieved slightly higher classification accuracy, MobileNetV3Small provides a more favorable balance between predictive performance and computational efficiency. Therefore, MobileNet may be preferable in environments prioritizing maximum diagnostic accuracy, whereas MobileNetV3Small is better suited for low-resource or real-time deployment scenarios.

To further illustrate the deployment characteristics of the evaluated models, Figure 6 presents the efficiency profiles of MobileNet and MobileNetV3Small.

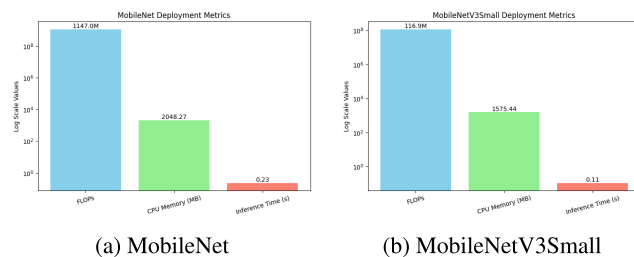


FIGURE 6. Deployment efficiency metrics of MobileNet and MobileNetV3Small.

3) MODEL SELECTION FOR SEMI-SUPERVISED LEARNING

Although MobileNet achieved the highest classification accuracy, the performance difference compared to MobileNetV3Small was relatively small (approximately 1–2%). However, MobileNetV3Small demonstrated significantly better computational efficiency. Considering its

competitive accuracy and significantly lower computational cost, MobileNetV3Small was selected as the backbone for subsequent semi-supervised learning experiments.

4) TRAINING PERFORMANCE OF MOBILENETV3SMALL

Figure 7 illustrates the training and validation accuracy and loss curves for MobileNetV3Small in both binary and multiclass classification tasks. Overall, the curves indicate stable convergence: accuracy steadily improves while loss decreases over epochs, with minimal overfitting in the binary task.

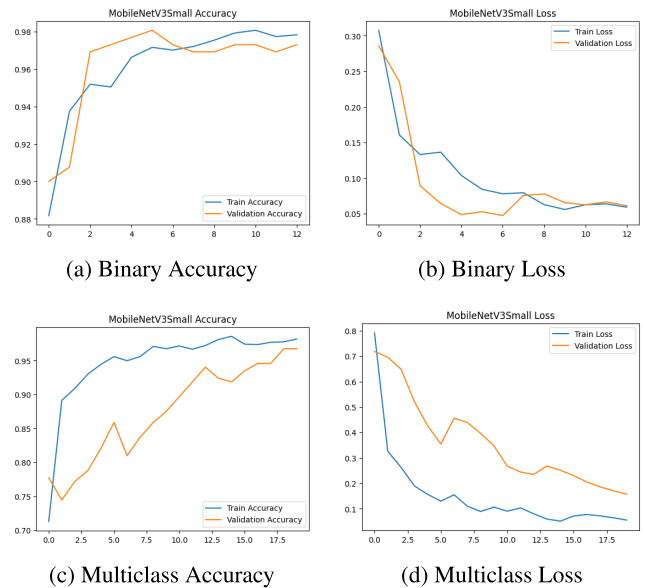


FIGURE 7. Training and validation accuracy and loss curves of MobileNetV3Small for binary and multiclass classification tasks.

For the multiclass task, training accuracy rises quickly during early epochs and stabilizes near 0.98, whereas validation accuracy fluctuates slightly before settling around 0.96. Similarly, training loss decreases consistently, while validation loss shows moderate variability. This behavior is expected due to the higher complexity of multiclass classification and similarities among pneumonia, COVID-19, and tuberculosis radiographic patterns. These results confirm that MobileNetV3Small effectively learns discriminative features across both tasks, providing strong generalization for subsequent semi-supervised experiments.

5) CROSS-VALIDATION PERFORMANCE

To assess model robustness, 3-fold cross-validation was conducted for both binary and multiclass tasks (Table 14). The fold-wise accuracies show that the binary task achieves consistently high performance with low standard deviation (0.76), while the multiclass task exhibits slightly higher variance (1.17) due to increased classification complexity.

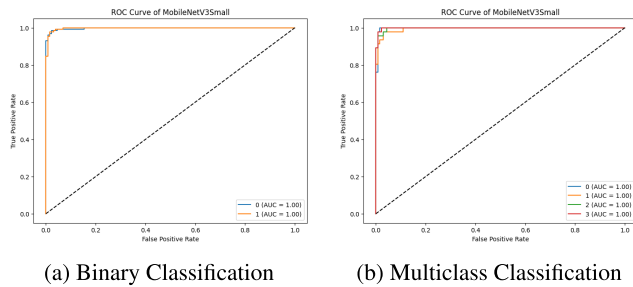
Overall, the cross-validation results reinforce that MobileNetV3Small delivers stable and reliable performance across both binary and multiclass chest X-ray classification tasks.

TABLE 14. 3-Fold cross-validation results for binary and multiclass classification.

Task	Fold 1	Fold 2	Fold 3	Mean	Std
Binary	96.54	97.46	95.61	96.54	0.76
Multiclass	96.42	93.64	94.45	94.84	1.17

6) ROC CURVE AND AUC ANALYSIS

ROC curves (Fig 8), were used to evaluate the discriminative ability of MobileNetV3Small, with AUC serving as the primary performance metric. In the binary task (Normal vs. Pneumonia), the ROC curve closely approaches the upper-left corner, indicating near-perfect separability with AUC close to 1.00. For the multiclass setting (COVID-19, Normal, Pneumonia, Tuberculosis), one-vs-rest ROC analysis shows consistently high AUC values across all classes, with only slight reductions in visually similar categories due to inter-class overlap. Overall, the high AUC values confirm strong class separability and robust generalization of the model across both tasks.

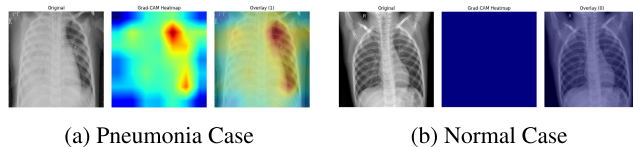
**FIGURE 8. ROC curves of MobileNetV3Small for (a) binary and (b) multiclass chest X-ray classification tasks.**

7) EXPLAINABILITY ANALYSIS USING GRAD-CAM

To enhance the interpretability of model predictions, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied. Grad-CAM generates heatmaps that highlight image regions that contribute most strongly to the predicted class, where warmer colors indicate stronger model attention.

a: BINARY CLASSIFICATION

Figure 9 presents Grad-CAM visualizations for the binary classification task (Normal vs. Pneumonia).

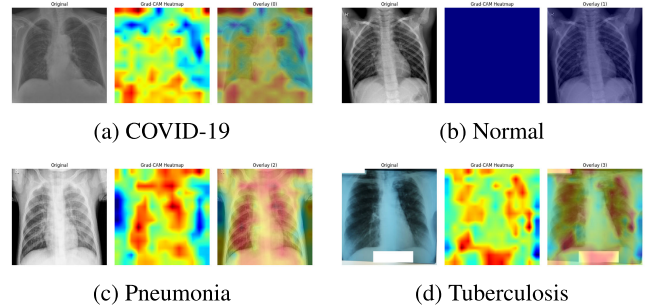
**FIGURE 9. Grad-CAM visualizations for binary chest X-ray classification.**

For the Pneumonia case, MobileNetV3Small predicted the positive class with a confidence of 0.989. The Grad-CAM heatmap highlights localized lung regions consistent with pathological changes, indicating that the model focuses on

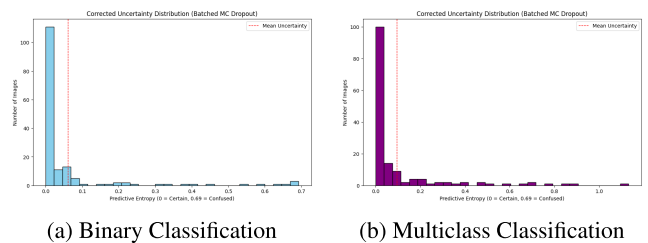
clinically relevant areas. In contrast, for the Normal case, the model predicted the negative class with a confidence of 0.998, and the Grad-CAM visualization shows minimal activation across the lung fields, suggesting the absence of abnormal patterns. These observations demonstrate that the model effectively differentiates between diseased and healthy lungs by focusing on meaningful anatomical structures.

b: MULTICLASS CLASSIFICATION

Grad-CAM was also applied to the multiclass classification task (Figure 10).

**FIGURE 10. Grad-CAM visualizations for multiclass chest X-ray classification.**

The model produced high-confidence predictions for all classes, including COVID-19 (0.911), Pneumonia (1.000), Tuberculosis (1.000), and Normal (0.999). The Grad-CAM visualizations indicate that strong activations correspond to lung regions affected by each respective disease, whereas the Normal class shows minimal activation. These results confirm that the network consistently attends to clinically relevant anatomical regions across all evaluated conditions. Importantly, the visualizations confirm that the model focuses on clinically relevant lung regions when identifying pneumonia, while distinguishing it from other respiratory diseases such as COVID-19 and tuberculosis. By linking model predictions with the spatial localization of disease-related features, Grad-CAM provides interpretable explanations that enhance trust and support the use of AI-assisted diagnostic systems in clinical practice.

**FIGURE 11. Predictive entropy distributions using Monte Carlo Dropout for binary and multiclass chest X-ray classification.**

8) UNCERTAINTY ANALYSIS

To evaluate prediction reliability, uncertainty estimation was performed using Monte Carlo Dropout. For each test image,

50 stochastic forward passes were executed while keeping dropout layers active during inference. Predictive entropy computed from the averaged probability distribution was used as the uncertainty metric.

Figure 11 shows the predictive entropy distributions for both binary and multiclass classification tasks. In the binary setting, most predictions are concentrated in the low-entropy region, indicating high confidence in distinguishing between Normal and Pneumonia cases. The model achieved an average entropy of 0.0613, with values ranging from 0.0000 to 0.6924. For multiclass classification, the entropy distribution is slightly broader due to the increased complexity of differentiating COVID-19, Normal, Pneumonia, and Tuberculosis cases. The model achieved an average entropy of 0.0952, with entropy values ranging from 0.0000 to 1.1325. Overall, only a small number of samples exhibit high uncertainty, suggesting that the proposed framework generates reliable predictions while effectively identifying challenging or ambiguous cases that may require additional expert review.

B. SEMI-SUPERVISED LEARNING RESULTS

In medical imaging, labeled data is often scarce due to the high cost of expert annotation. Semi-supervised learning (SSL) techniques leverage unlabeled data through pseudo-labeling and consistency regularization, improving model performance even with limited labeled samples. In this study, three state-of-the-art SSL methods—FixMatch, FlexMatch, and FreeMatch were evaluated on both binary classification (Normal vs. Pneumonia) and multiclass classification (COVID-19, Normal, Pneumonia, Tuberculosis). Performance metrics included accuracy, precision, recall, and F1-score, providing a comprehensive assessment under varying labeled data proportions.

1) BINARY CLASSIFICATION RESULTS

The comparative results of FreeMatch, FixMatch, and FlexMatch across 10%, 20%, and 30% labeled settings show that all methods achieve strong classification performance; however, their stability and robustness differ under low-label conditions.

At the 10% labeled setting, FreeMatch achieves competitive performance (Accuracy = 0.95, F1-score = 0.95) with more stable training behavior compared to FixMatch and FlexMatch. FlexMatch shows higher variance, while FixMatch is relatively stable but slightly less adaptive in low-data scenarios. This indicates that FreeMatch is more effective in handling extreme label scarcity due to its adaptive thresholding mechanism for pseudo-label selection.

At the 20% setting, FreeMatch demonstrates the most stable improvement (0.9577 ± 0.0113), showing better utilization of unlabeled data compared to the other methods. The Mean $\pm$ Std values denote the mean and standard deviation of the best validation accuracies across three independent runs. At 30%, all methods converge to similar performance, indicating reduced impact of semi-supervised strategies as labeled data increases. (Table 15)

TABLE 15. Overall performance comparison of FreeMatch, FixMatch, and flexmatch (10%–30% labeled data).

Method	Setting	Acc.	Prec.	Rec.	F1	Val. Acc. (Mean $\pm$ Std)
FreeMatch	10%	0.95	0.95	0.95	0.95	0.8667 ± 0.1151
	20%	0.97	0.97	0.97	0.97	0.9577 ± 0.0113
	30%	0.95	0.95	0.95	0.95	0.9500 ± 0.0220
FixMatch	10%	0.95	0.95	0.95	0.95	0.8449 ± 0.0846
	20%	0.95	0.95	0.95	0.95	0.9282 ± 0.0271
	30%	0.97	0.97	0.97	0.97	0.9474 ± 0.0127
FlexMatch	10%	0.95	0.95	0.95	0.95	0.8192 ± 0.1151
	20%	0.95	0.95	0.95	0.95	0.8769 ± 0.0724
	30%	0.97	0.97	0.97	0.97	0.9436 ± 0.0184

Class-wise results further show that FreeMatch maintains balanced precision and recall across both classes, indicating consistent generalization without bias toward any class. (Table 16)

TABLE 16. Class-wise performance comparison of FreeMatch, FixMatch, and flexmatch (10%–30% labeled data).

Model	Setting	Class	Precision	Recall	F1-score
FreeMatch	10%	0	0.98	0.93	0.95
		1	0.93	0.98	0.95
		Macro Avg	0.95	0.95	0.95
		Weighted Avg	0.95	0.95	0.95
	20%	0	0.96	0.97	0.97
		1	0.97	0.96	0.97
		Macro Avg	0.97	0.97	0.97
		Weighted Avg	0.97	0.97	0.97
	30%	0	0.90	1.00	0.95
		1	1.00	0.89	0.94
		Macro Avg	0.95	0.95	0.95
		Weighted Avg	0.95	0.95	0.95
FixMatch	10%	0	0.94	0.95	0.95
		1	0.95	0.94	0.95
		Macro Avg	0.95	0.95	0.95
		Weighted Avg	0.95	0.95	0.95
	20%	0	0.95	0.95	0.95
		1	0.95	0.95	0.95
		Macro Avg	0.95	0.95	0.95
		Weighted Avg	0.95	0.95	0.95
	30%	0	0.99	0.94	0.96
		1	0.94	0.99	0.97
		Macro Avg	0.97	0.97	0.97
		Weighted Avg	0.97	0.97	0.97
FlexMatch	10%	0	0.91	0.99	0.95
		1	0.99	0.90	0.94
		Macro Avg	0.95	0.95	0.95
		Weighted Avg	0.95	0.95	0.95
	20%	0	0.91	0.99	0.95
		1	0.99	0.90	0.94
		Macro Avg	0.95	0.95	0.95
		Weighted Avg	0.95	0.95	0.95
	30%	0	0.98	0.96	0.97
		1	0.96	0.98	0.97
		Macro Avg	0.97	0.97	0.97
		Weighted Avg	0.97	0.97	0.97

To further validate robustness, FreeMatch was also evaluated on the full dataset, confirming stable performance across different labeling ratios. Overall, FreeMatch is the most reliable method in low-label scenarios due to its stability, balanced class performance, and lower variance across runs. (Table 17)

2) MULTICLASS CLASSIFICATION RESULTS

Multiclass classification introduces additional challenges due to the need to distinguish among four respiratory conditions. From the overall and class-wise evaluation,

TABLE 17. FreeMatch performance on full dataset (10%–30% labeled data).

Setting	Acc.	Prec.	Rec.	F1	Val. Acc. (Mean $\pm$ Std)
10%	0.91	0.92	0.91	0.91	0.9479 ± 0.0029
20%	0.92	0.93	0.92	0.92	0.9438 ± 0.0051
30%	0.90	0.91	0.90	0.90	0.9604 ± 0.0029

FreeMatch demonstrates the most consistent and stable performance across all labeling ratios (10%–30%). In particular, FreeMatch achieves higher or comparable accuracy and F1-score than both FlexMatch and FixMatch, while maintaining lower variance across multiple runs, indicating better training stability under limited supervision.

At the 10% labeled setting, FreeMatch significantly outperforms FixMatch in all evaluation metrics, highlighting its robustness in extremely low-label regimes. FlexMatch shows competitive performance but suffers from class imbalance sensitivity, particularly in minority-class recall, where performance drops noticeably compared to FreeMatch. As the labeled data increases (20%–30%), all methods improve; however, FreeMatch maintains more balanced precision and recall across all classes, especially in difficult classes where FixMatch and FlexMatch exhibit higher variance in class-wise recall. This indicates that FreeMatch produces more reliable pseudo-labels and reduces confirmation bias during training. Overall, FreeMatch provides the best trade-off between accuracy, stability (low standard deviation), and class-wise balance. (Table 18, 19)

TABLE 18. Overall performance comparison of FreeMatch, FlexMatch, and fixmatch (multiclass, 10%–30% labeled data).

Method	Setting	Acc.	Prec.	Rec.	F1	Val. Acc. (Mean $\pm$ Std)
FreeMatch	10%	0.88	0.89	0.88	0.88	0.8271 ± 0.0741
	20%	0.93	0.93	0.93	0.92	0.9135 ± 0.0367
	30%	0.97	0.97	0.97	0.97	0.9323 ± 0.0307
FlexMatch	10%	0.85	0.87	0.85	0.85	0.7958 ± 0.0892
	20%	0.94	0.94	0.94	0.94	0.8813 ± 0.0687
	30%	0.93	0.93	0.93	0.93	0.9062 ± 0.0381
FixMatch	10%	0.77	0.86	0.74	0.75	0.7719 ± 0.0575
	20%	0.88	0.93	0.92	0.92	0.8813 ± 0.0873
	30%	0.94	0.93	0.92	0.92	0.9417 ± 0.0362

The training process remained stable across different labeled data ratios, as evidenced by consistent performance and low variance.

3) ABLATION STUDY

Table 20 presents the ablation study of the FreeMatch framework under different labeled data ratios for binary and multi-class chest X-ray classification. The results demonstrate consistent performance across both settings, even with limited labeled samples, indicating strong semi-supervised learning capability. The results show that FreeMatch maintains stable precision and recall across both binary and multi-class settings, even with only 20% and 30% labeled data, respectively. Slight variations in class-wise performance indicate sensitivity to inter-class similarity in multi-class

TABLE 19. Class-wise performance comparison of FreeMatch, FlexMatch, and fixmatch (multiclass).

Method	Setting	Class	Precision	Recall	F1-score
FreeMatch	10%	0	0.86	0.86	0.86
		1	0.78	1.00	0.87
		2	0.92	0.85	0.88
		3	1.00	0.79	0.88
		Macro Avg	0.89	0.88	0.88
		Weighted Avg	0.89	0.88	0.88
	20%	0	0.97	0.85	0.91
		1	0.87	0.97	0.92
		2	0.89	0.90	0.89
		3	0.99	0.97	0.98
		Macro Avg	0.93	0.93	0.92
		Weighted Avg	0.93	0.93	0.92
	30%	0	0.96	0.95	0.96
		1	0.96	1.00	0.98
		2	0.97	0.97	0.97
		3	0.97	0.95	0.96
		Macro Avg	0.97	0.97	0.97
		Weighted Avg	0.97	0.97	0.97
FlexMatch	10%	0	0.98	0.61	0.75
		1	0.81	0.97	0.89
		2	0.78	0.91	0.84
		3	0.91	0.91	0.91
		Macro Avg	0.87	0.85	0.85
		Weighted Avg	0.87	0.85	0.85
	20%	0	0.96	0.90	0.93
		1	0.92	0.96	0.94
		2	0.93	0.93	0.93
		3	0.96	0.97	0.97
		Macro Avg	0.94	0.94	0.94
		Weighted Avg	0.94	0.94	0.94
	30%	0	1.00	0.88	0.93
		1	0.96	0.86	0.91
		2	0.85	0.97	0.91
		3	0.93	1.00	0.96
		Macro Avg	0.93	0.93	0.93
		Weighted Avg	0.93	0.93	0.93
FixMatch	10%	0	0.97	0.70	0.81
		1	0.95	0.53	0.68
		2	0.51	1.00	0.67
		3	1.00	0.75	0.86
		Macro Avg	0.86	0.74	0.75
		Weighted Avg	0.86	0.74	0.75
	20%	0	1.00	0.81	0.90
		1	0.93	0.89	0.91
		2	0.84	0.99	0.91
		3	0.94	1.00	0.97
		Macro Avg	0.93	0.92	0.92
		Weighted Avg	0.93	0.92	0.92
	30%	0	0.99	0.93	0.95
		1	0.96	0.95	0.96
		2	0.94	0.97	0.96
		3	0.95	0.99	0.97
		Macro Avg	0.96	0.96	0.96
		Weighted Avg	0.96	0.96	0.96

classification. Overall, the model demonstrates strong robustness under low-label training conditions.

4) CONFUSION MATRIX ANALYSIS

Confusion matrices provide a class-wise evaluation of prediction correctness for both binary and multiclass tasks. Figure 12 summarizes the results obtained using the proposed FreeMatch framework with a MobileNetV3Small backbone.

TABLE 20. Ablation study of FreeMatch under best label ratios.

Setting	Label %	Class	Precision	Recall	F1-score
Binary	20%	Normal	0.96	0.97	0.97
		Pneumonia	0.97	0.96	0.97
Multi-class	30%	Normal	0.96	1.00	0.98
		Pneumonia	0.97	0.97	0.97

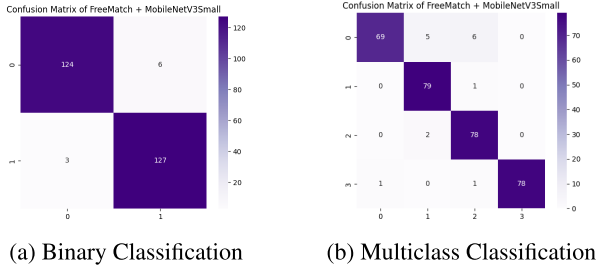


FIGURE 12. Confusion matrices of the proposed freematch framework.

For binary classification, the model shows strong diagonal dominance with 124 correct predictions for Class 0 and 127 for Class 1, with minimal misclassifications. This indicates balanced sensitivity across both categories. For multiclass classification, the model achieves consistently high correct predictions across all four classes. Class 1 shows the strongest performance, while minor confusion is observed in Class 0 due to visual similarity with adjacent classes. Overall, the results confirm strong class separability and robust feature learning.

5) ACCURACY AND LOSS CURVE ANALYSIS

Figure 13 shows the training and validation accuracy/loss curves of the proposed FreeMatch framework for binary and multiclass tasks over three runs. For binary classification, the training accuracy has been high and the validation accuracy has been steadily improving. The loss curves showed a stable convergence, with the training and validation losses being very close to each other. Similarly, for multiclass classification, the model has achieved good training and validation performance with stable loss trends across runs. No significant overfitting was found and only small fluctuations were observed. Overall, the curves verify the robustness and generalization ability of the proposed framework.

6) ROC CURVE ANALYSIS

ROC analysis evaluates the discriminative ability of the proposed framework (Figure 14). The model achieves an AUC of 1.00 for binary and multiclass classification, suggesting near perfect class separability with few false positives. The ROC curves are very close to the top-left corner which indicates that the proposed method can discriminate well with limited labeled data.

7) MONTE CARLO DROPOUT UNCERTAINTY ANALYSIS

Monte Carlo Dropout (MC Dropout) is applied to estimate the predictive uncertainty through entropy for both binary

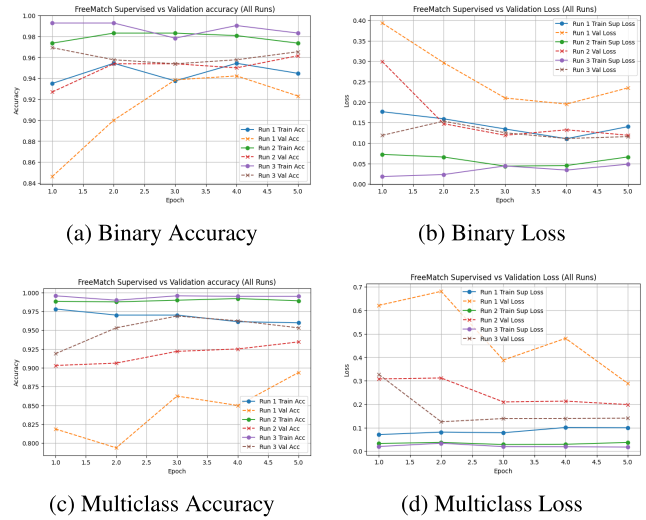


FIGURE 13. Training and validation accuracy/loss curves of the proposed FreeMatch framework across three independent runs.

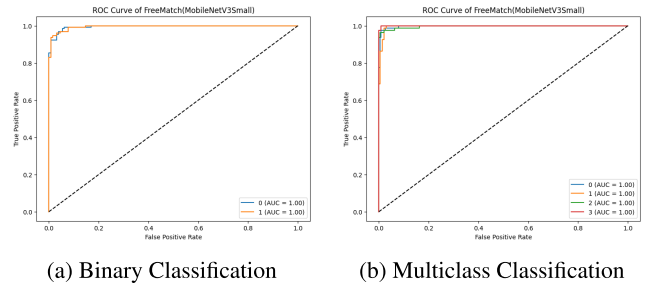


FIGURE 14. ROC curves of the proposed FreeMatch framework.

and multiclass classification using the MobileNetV3Small in the FreeMatch framework. Both tasks exhibit strongly left-skewed distributions (Fig. 15) and the majority of samples have near-zero entropy, meaning high model confidence. The binary task has a mean uncertainty of 0.1384, while the multiclass task has an even lower mean of 0.0760 with only a small fraction of high uncertainty samples.

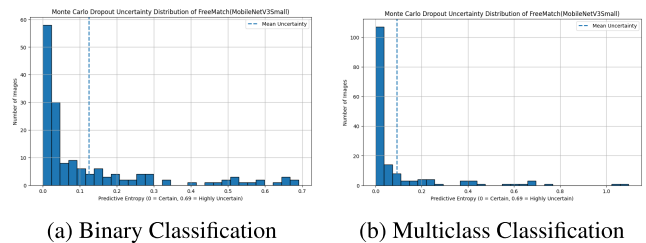


FIGURE 15. MC Dropout uncertainty distribution.

8) MC DROPOUT THRESHOLD AND CALIBRATION ANALYSIS

Using an entropy threshold of $\tau = 0.3$, predictions were divided into certain and uncertain groups (Fig. 16). For the binary task, 17.58% of samples were flagged as uncertain, with accuracy dropping from 98.48% (certain) to 82.14%

(uncertain). In the multiclass case, accuracy decreased from 99.3% to 85.7%, showing a consistent performance gap.

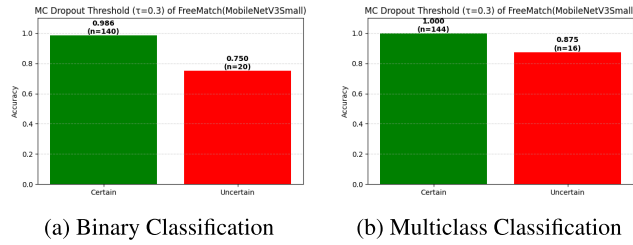


FIGURE 16. MC Dropout threshold-based uncertainty analysis.

A clear positive relationship between predictive entropy and misclassification rate is observed in Fig. 17, confirming that higher uncertainty corresponds to higher prediction error. This indicates that MC Dropout provides well-calibrated uncertainty estimates.

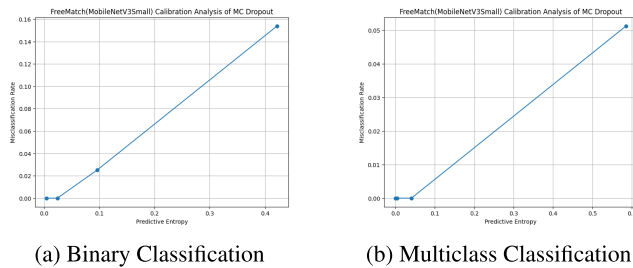


FIGURE 17. Predictive entropy vs misclassification rate.

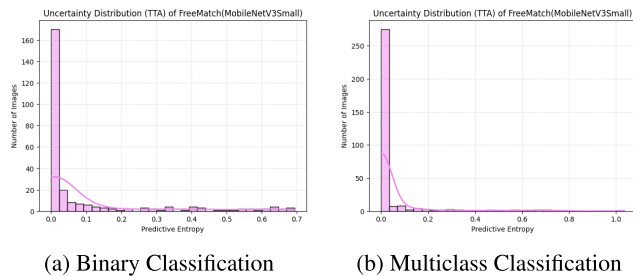


FIGURE 18. Predictive entropy distributions using test-time augmentation (TTA) with FreeMatch and MobileNetV3Small.

9) TEST-TIME AUGMENTATION ANALYSIS

As shown in Fig. 18, both tasks exhibit strongly left-skewed entropy distributions, indicating high model confidence. For binary classification, most samples lie in the 0.0–0.05 entropy range, with a TTA accuracy of 0.9500. A small tail extends up to 0.7, representing uncertain cases. For multiclass classification, over 250 samples are concentrated near zero entropy, with a TTA accuracy of 0.9812. Only a few samples show mild uncertainty (0.1–0.2), indicating stable predictions under augmentation.

Figure 19 summarizes the impact of TTA on classification performance. In the binary setting, performance slightly

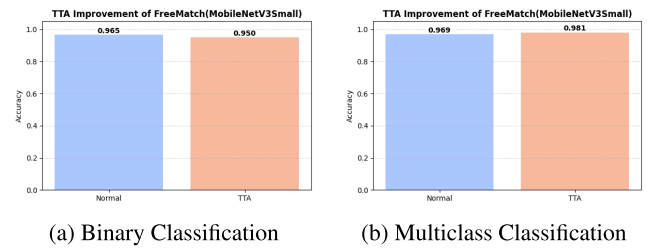


FIGURE 19. Performance comparison between normal inference and test-time augmentation (TTA).

decreased from 96.5% to 95%, indicating that the baseline model was already highly stable. In contrast, multiclass classification improved from 96.9% to 98.1%, showing that TTA helps reduce ambiguity in more complex classification scenarios.

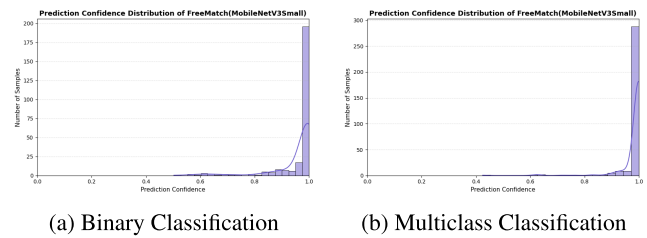


FIGURE 20. Prediction confidence distributions of FreeMatch with MobileNetV3Small for binary and multiclass classification.

10) PREDICTION CONFIDENCE DISTRIBUTION

Figure 20 presents the prediction confidence distributions of FreeMatch with MobileNetV3Small. In both tasks, prediction confidence is heavily concentrated near 1.0, indicating strong confidence for most samples. The distributions are consistent with the low calibration error values observed in later analysis, suggesting reliable probabilistic predictions.

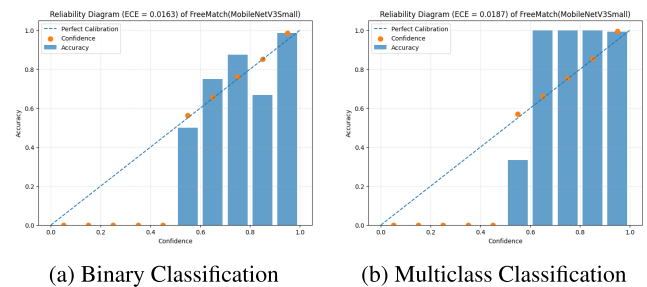


FIGURE 21. Reliability diagrams for binary and multiclass classification using FreeMatch and MobileNetV3Small.

11) RELIABILITY AND CALIBRATION ANALYSIS

Calibration performance was evaluated using reliability diagrams and Expected Calibration Error (ECE), which measure the alignment between predicted confidence and empirical accuracy. Figure 21 presents the reliability diagrams for binary and multiclass classification. The binary

model achieved an ECE of 0.0163, while the multiclass model achieved an ECE of 0.0187, demonstrating strong calibration quality across both tasks. Although minor deviations appear in intermediate confidence regions, overall results show strong agreement between predicted confidence and actual accuracy, supporting the reliability of the proposed framework for clinical decision support.

TABLE 21. Comparison with representative prior studies.

Study	Main Result
CheXNet [17]	F1-score 0.435
COVID-Net [6]	93.3% accuracy
Ozturk et al. [5]	98.08% binary; 87.02% multiclass accuracy
Srivastava et al. [7]	97.5% accuracy
Proposed Framework	97% accuracy (20% labeled, binary and 30% labeled, multiclass) using adaptive pseudo-labeling

V. DISCUSSION

The experimental results demonstrate that the proposed framework achieves strong and consistent performance across both supervised and semi-supervised chest X-ray classification tasks while maintaining computational efficiency through MobileNetV3Small. As summarized in Table 21, the proposed method achieves competitive or superior performance compared to prior supervised approaches such as CheXNet [17], COVID-Net [6], and other multi-class CNN-based models [7], while relying on substantially fewer labeled samples. In particular, it achieves 97% accuracy under both a 20% labeled binary setting and 30% labeled multiclass setting, demonstrating its suitability for annotation-limited clinical environments.

The observed performance improvements are associated with adaptive pseudo-label filtering and class-aware confidence balancing, which help reduce confirmation bias and improve label quality under low-label regimes. While the inclusion of COVID-19 and tuberculosis as clinically related classes may contribute to a richer feature representation for respiratory disease learning, this effect is not explicitly isolated in the current experimental design and should be interpreted as a supporting hypothesis rather than a direct causal finding. Compared with conventional semi-supervised learning approaches such as FixMatch, FlexMatch, and FreeMatch, which are largely validated on natural image benchmarks, the proposed framework demonstrates improved stability in multiclass thoracic disease classification, indicating that domain-specific adaptation of pseudo-labeling strategies is critical when transferring SSL methods to medical imaging tasks. However, limitations remain in distinguishing visually similar classes such as pneumonia, COVID-19, and tuberculosis, where overlapping radiographic patterns increase pseudo-label uncertainty and misclassification rates. Uncertainty analysis further indicates that most errors occur in high-entropy samples, suggesting reduced confidence reliability in ambiguous cases despite overall strong calibration. Overall, while semi-supervised learning significantly reduces annotation

requirements, further improvements in uncertainty modeling and inter-class separability are necessary for robust real-world clinical deployment. A limitation of this study is that all experiments were conducted on publicly available benchmark datasets. Although consistent performance was observed across binary and multiclass settings, external validation using multi-center or real-world clinical datasets is necessary to further assess the generalizability and robustness of the proposed framework. A limitation of this study is that there is no external validation on multi-center clinical datasets to evaluate generalizability. In addition, the auxiliary COVID-19 and tuberculosis classes were not isolated by binary-only ablation, and their contribution should be considered as a hypothesis based on indirect evidence. Future work will include controlled ablation and external testing to further validate the framework.

VI. CONCLUSION AND FUTURE WORK

This study presents a unified and computationally efficient framework for chest X-ray classification using lightweight MobileNet architectures integrated with semi-supervised learning, explainability, and uncertainty estimation. The proposed approach achieves strong performance in both binary and multiclass respiratory disease classification while significantly reducing reliance on fully labeled data. Among the evaluated architectures, MobileNetV3Small provides an effective balance between accuracy and computational efficiency, making it suitable for resource-constrained medical environments. The framework demonstrates the effectiveness of semi-supervised learning under limited annotation settings, with FreeMatch showing improved stability compared to FixMatch and FlexMatch. In addition, explainability through Grad-CAM and uncertainty estimation through calibration analysis enhance the transparency and reliability of predictions, supporting its potential use in clinical decision-support systems. Despite these advantages, the current study is limited by evaluation on benchmark datasets that may not fully reflect real-world clinical variability. Future work will focus on validating the proposed framework on external, multi-institutional chest X-ray datasets collected from different hospitals and imaging devices to assess its generalizability across diverse patient populations and clinical environments. Additionally, prospective clinical evaluations in real-world healthcare settings could further establish its robustness and practical applicability.

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